

# Practice Geometry Final Test-Part I (Version 1)

Name  
Date  
Course

- I) Write your name, the date, and the course title (i.e. Geometry).
- II) Show all work.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) You may use a calculator for some problems on this test.
- V) Each problem is worth four points for a total of 100 points.

① If the measure of an angle is fourteen degrees more than the measure of its complement, what is the measure of the angle?

$$\begin{aligned}
 & \begin{cases} A = 14 + C \\ A + C = 90 \end{cases} \\
 & \rightarrow 14 + C + C = 90 \\
 & \quad 14 + 2C = 90 \\
 & \quad 2C = 76 \\
 & \quad C = 38^\circ \\
 & \quad A = 14 + 38 \\
 & \quad \boxed{A = 52^\circ}
 \end{aligned}$$

② What type of reasoning is used below?

Johnny used the Triangle Sum Theorem, the definition of angle measure, and the Angle-Angle Similarity Postulate to show that two triangles are similar.

Deductive

③ What word do we use for a conclusion generated by using a pattern in past events?

Conjecture

④ If we form a logical conclusion based on the assumption that the statements below are true, then what law of logic did we use?

- a) If Kary wins her case, then she will be promoted.
- b) Kary won her case yesterday.

The Law of Detachment

⑤ Predict the next symbol in the sequence below.



⑥ If the weather is warm, then it is summertime. Write a counterexample for the previous statement.

The weather is sometimes warm during the winter.

⑦ Match a, b, or c for parts i-iii below.

Original: If Don enters the tournament, then he will win first place.

\* Remember, these statements do not have to make sense.

i) If Don does not win first place, then he will not enter the tournament.

b

a) Converse

b) Contrapositive

ii) If Don wins first place, then he will enter the tournament.

a

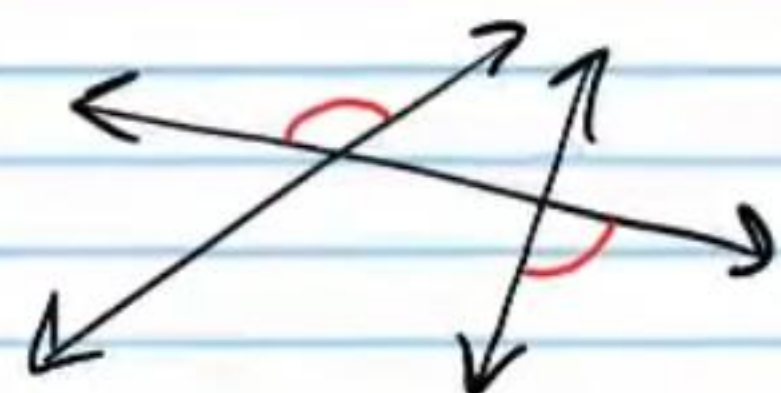
c) Inverse

iii) If Don does not enter the tournament, then he will not win first place.

c



iv) What is the name of the angle pair below?



Alternate Exterior Angles

8) Is it possible for the biconditional statement below to be true? Explain.

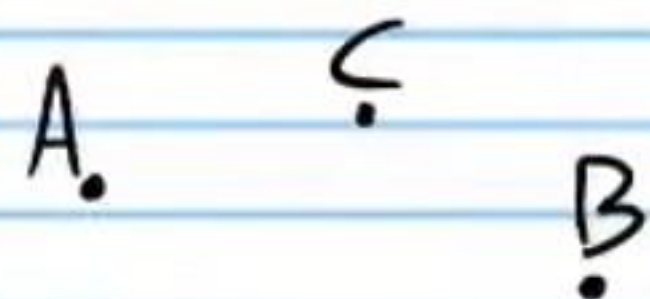
Laura will buy her mother a house if and only if she wins the lottery.

No. Buying her mother a house will not guarantee that she wins the lottery.

9i) What is formed by the intersection of two lines?

A point

ii) Are points A, B, and C coplanar?



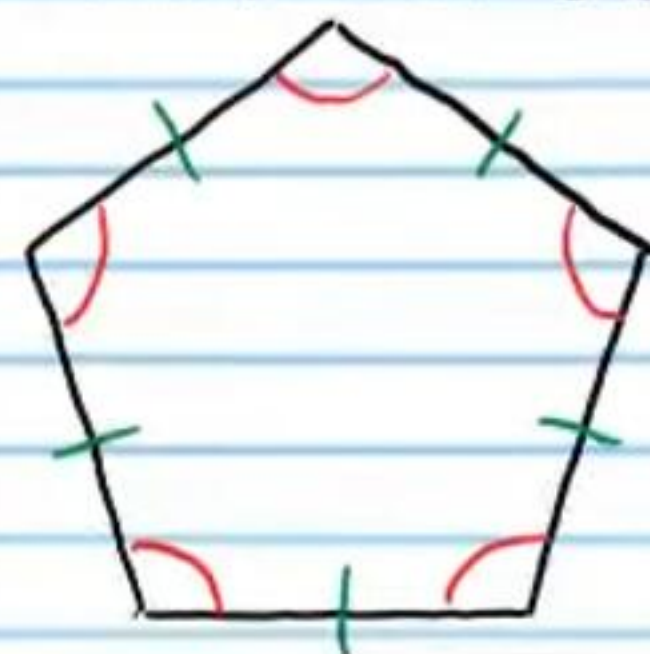
Yes. Any three points are coplanar.

10i) Name the following polygon by its number of sides.

Pentagon

ii) Is it equilateral?

Yes



iii) Is it equiangular? Yes

iv) Is it regular? Yes

11i) If the triangle below were classified by its sides, what word would be used to describe it?

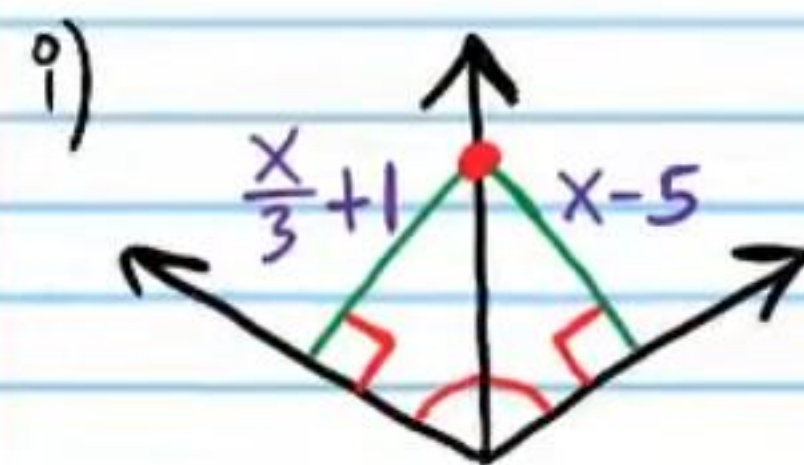


Scalene

ii) If the triangle above were classified by its angles, what word would be used to describe it?

Obtuse

12) Find  $x$  if possible.



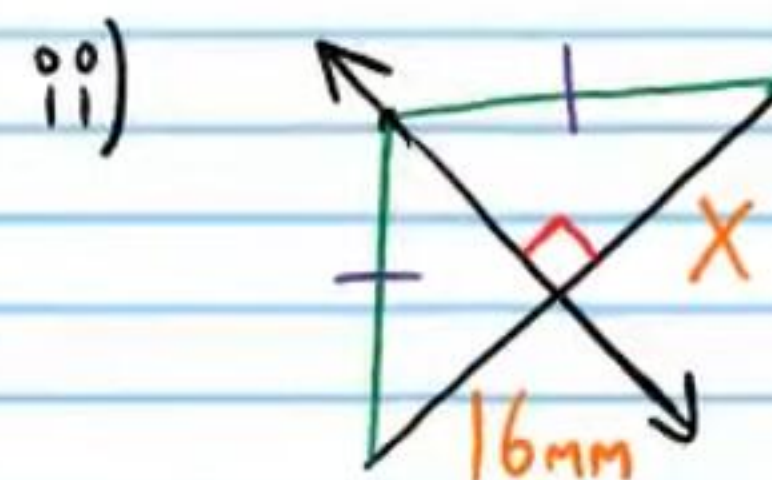
$$\frac{x}{3} + 1 = x - 5$$

$$6 = x - \frac{x}{3}$$

$$6 = \frac{2x}{3}$$

$$18 = 2x$$

$$9 = x$$

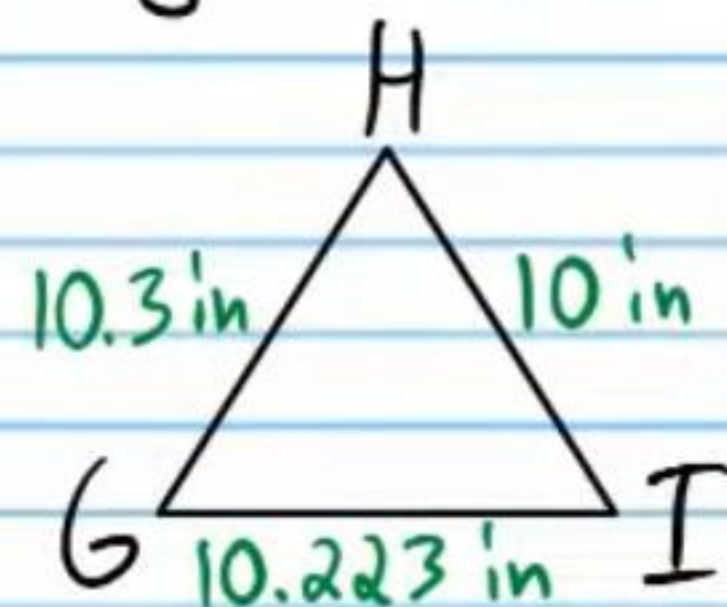


$$x = 16 \text{ mm}$$

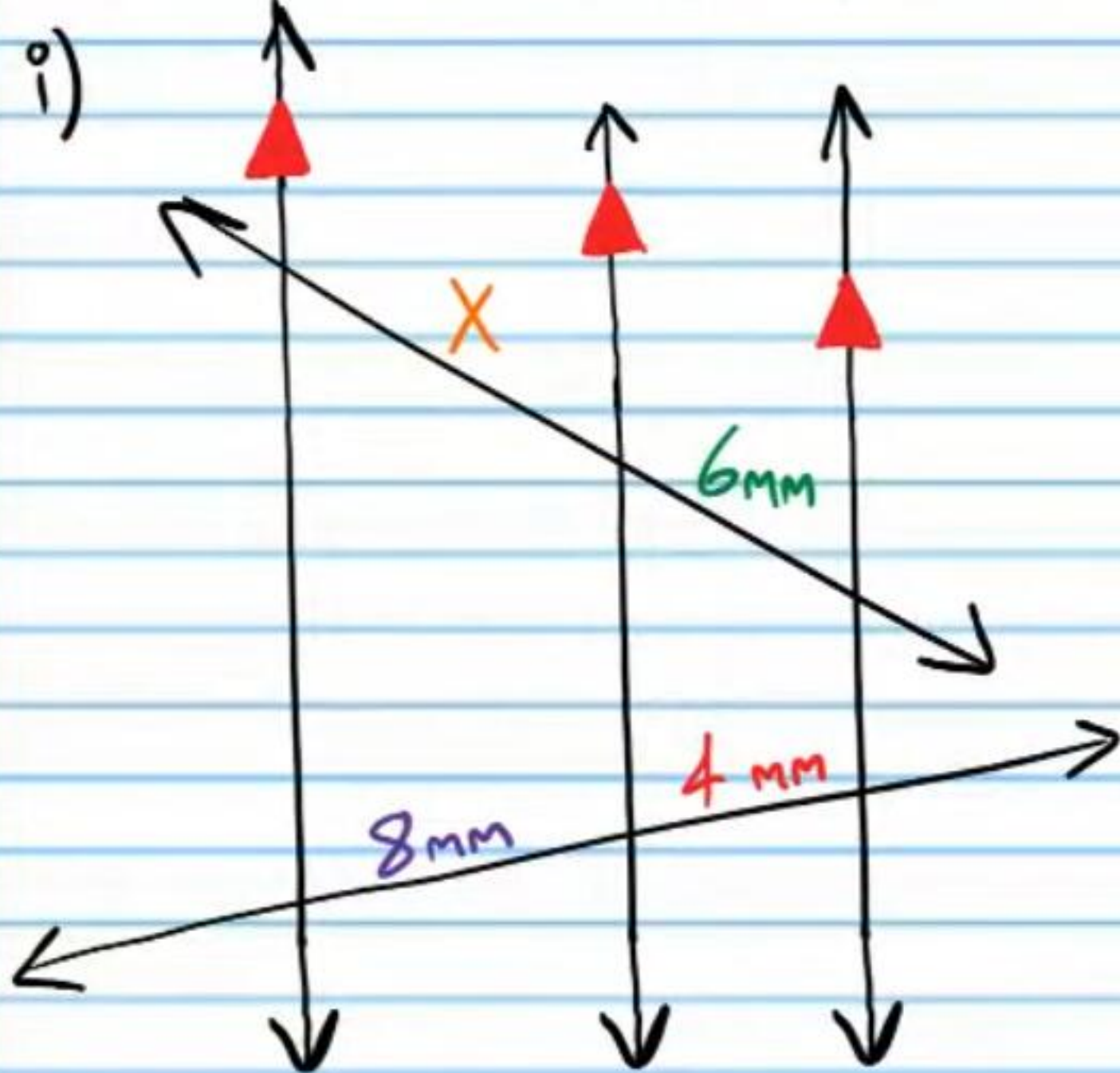


- ⑬ List the angles of the triangle below from least to greatest.

$\angle G, \angle H, \angle I$



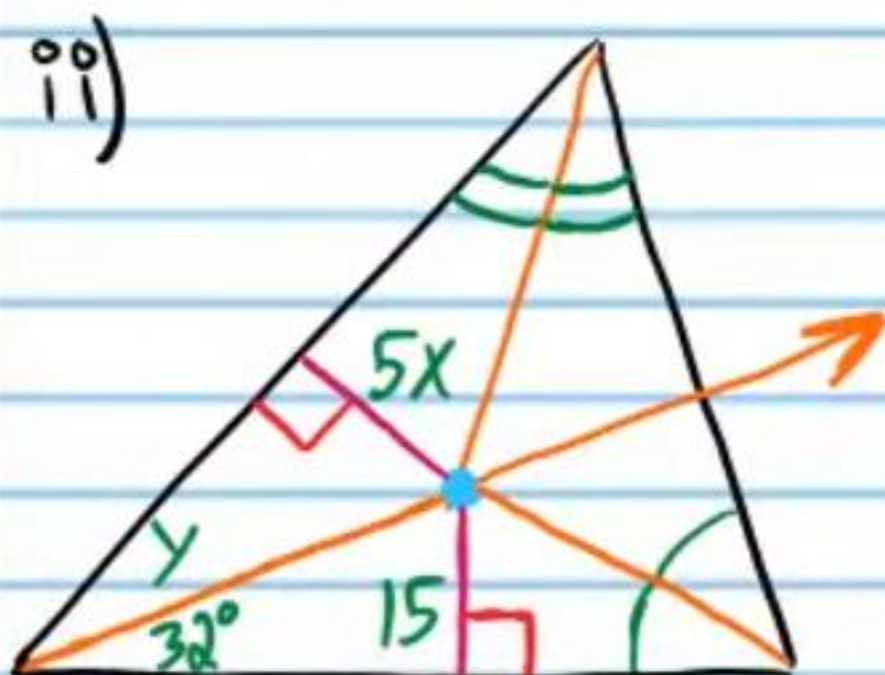
- ⑭ Find  $x$  and/or  $y$ .



$$\frac{x}{6} = \frac{8}{4}$$

$$\frac{x}{6} = 2$$

$$x = 12 \text{ mm}$$



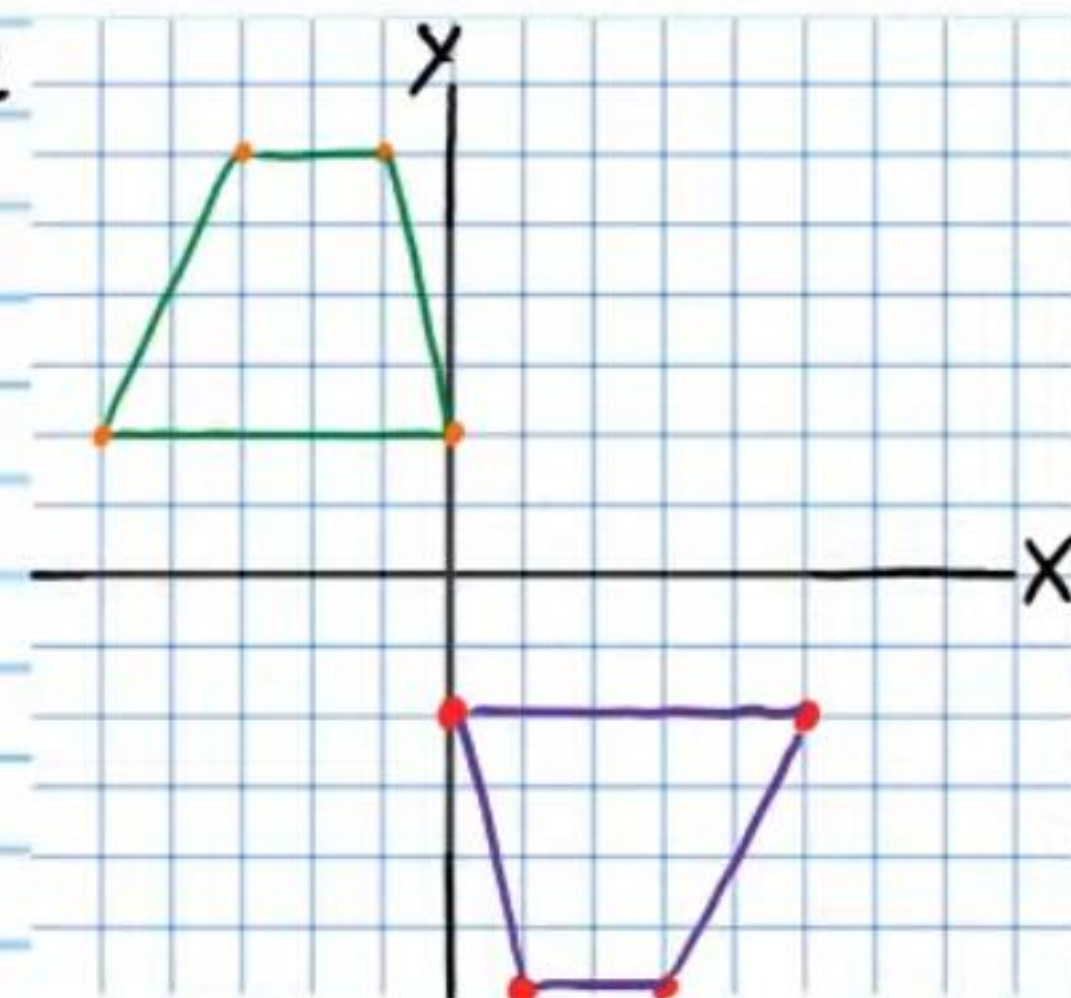
I)  $5x = 15$

$$x = 3$$

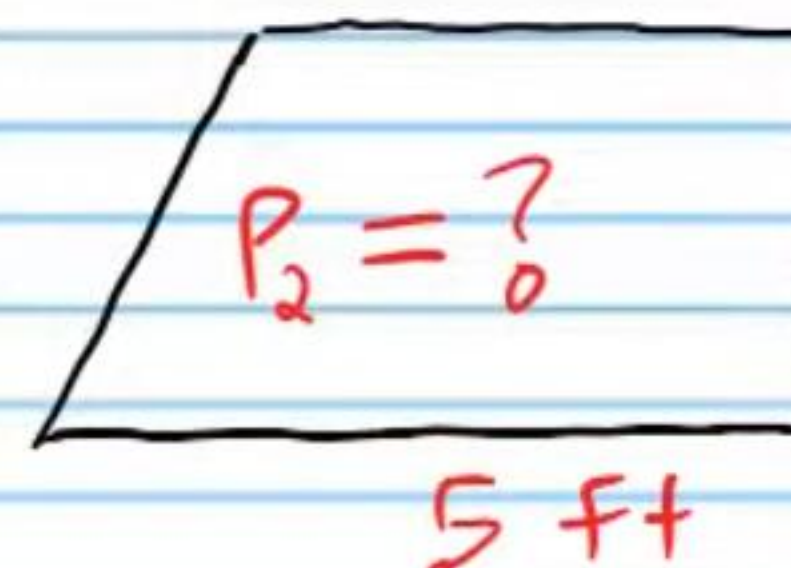
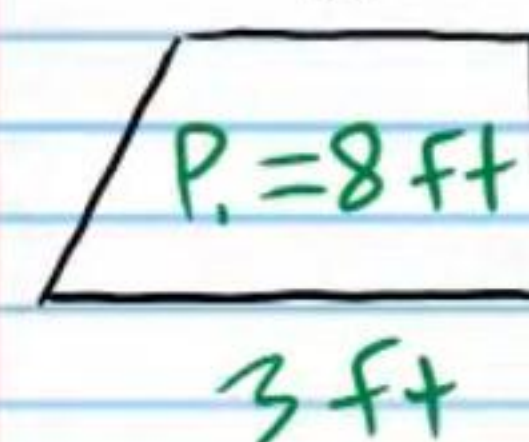
II)  $y = 32^\circ$

- ⑮ Perform the given transformation on the pre-image below and draw the image.

Rotate  $180^\circ$  Counterclockwise



- ⑯ If the polygons below are similar, find the missing perimeter.



$$\frac{P_2}{P_1} = \frac{5}{3}$$

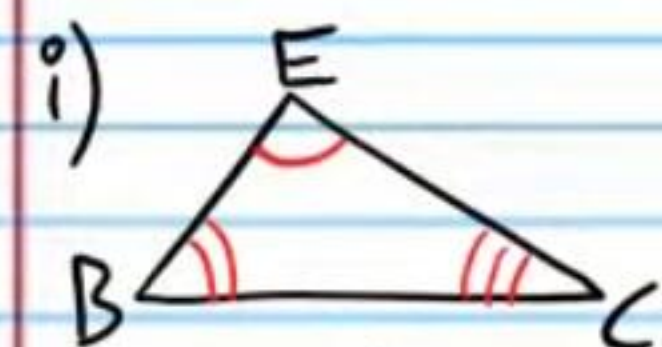
$$\frac{P_2}{8} = \frac{5}{3}$$

$$P_2 = \frac{40}{3}$$

$$P_2 = 13\frac{1}{3} \text{ ft}$$



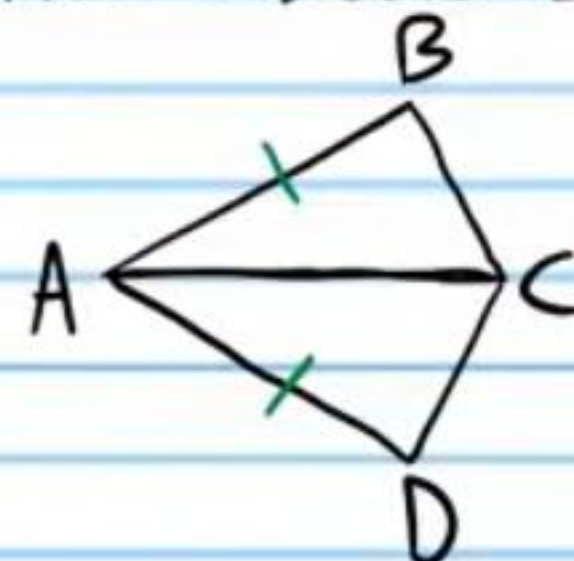
⑪ In the problems below, determine if the triangles are congruent. If they are congruent, write a proper congruence statement and the triangle postulate or theorem that proves that they are congruent. You do not have to write a formal proof (i.e. a two-column proof).



$\triangle CBE \cong$  ?

Not Enough Information

ii)  $\overline{AC}$  bisects  $\angle A$ .



$\triangle ABC \cong$  ?

$\triangle ABC \cong \triangle ADC$   
SAS

⑫ Is it possible to make a triangle with the side lengths below? Show the inequality or inequalities that demonstrate your answer.

9, 18, 27

$$9 + 18 > 27$$

$$27 > 27$$

X

$$18 + 27 > 9$$

$$45 > 9$$

✓

$$9 + 27 > 18$$

$$36 > 18$$

✓

No

⑬ If point D is the orthocenter of the triangle below, find the quantities listed.

a)  $m\angle DEA = 90^\circ$

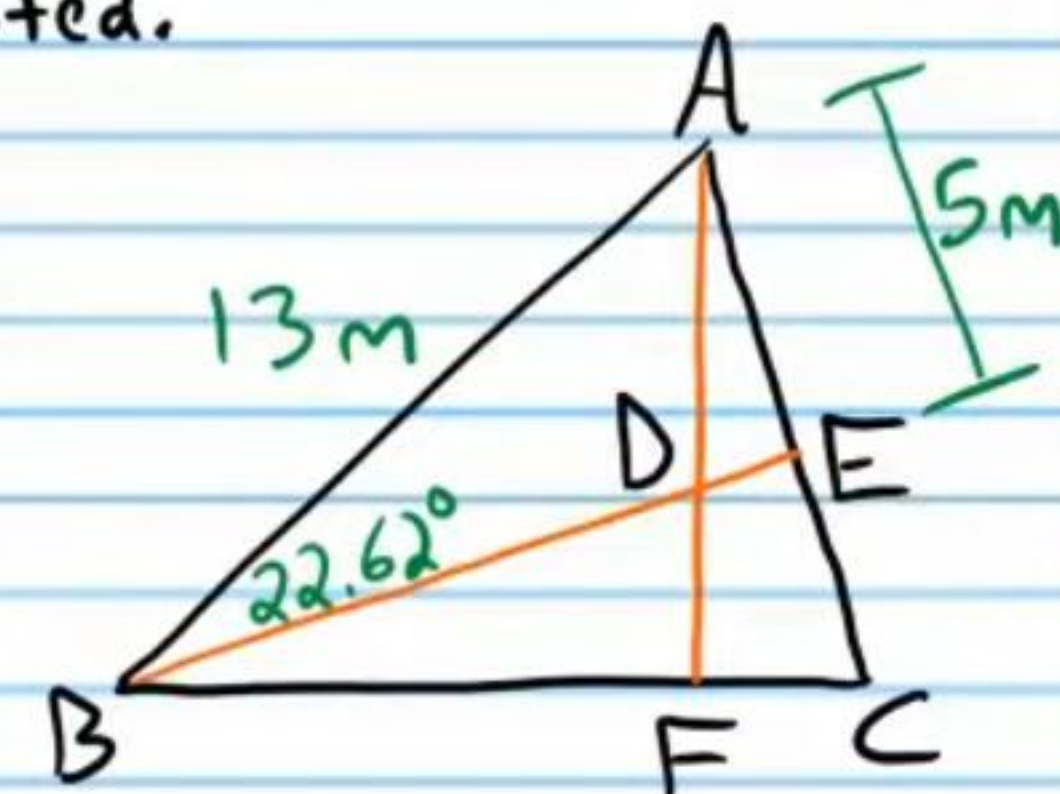
b)  $m\angle BFD = 90^\circ$

c)  $m\angle BAE$

$$m\angle BAE + 90 + 22.62 = 180$$

$$m\angle BAE + 112.62 = 180$$

$$m\angle BAE = 67.38^\circ$$



d) BE

$$5^2 + BE^2 = 13^2$$

$$25 + BE^2 = 169$$

$$BE^2 = 144$$

$$\sqrt{BE^2} = \pm \sqrt{144}$$

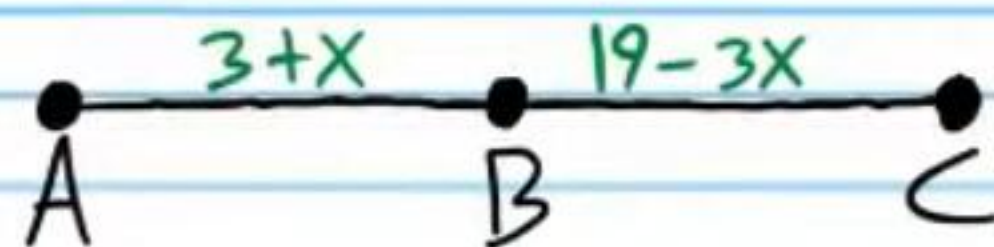
$$BE = \pm 12$$

$$BE = 12m$$



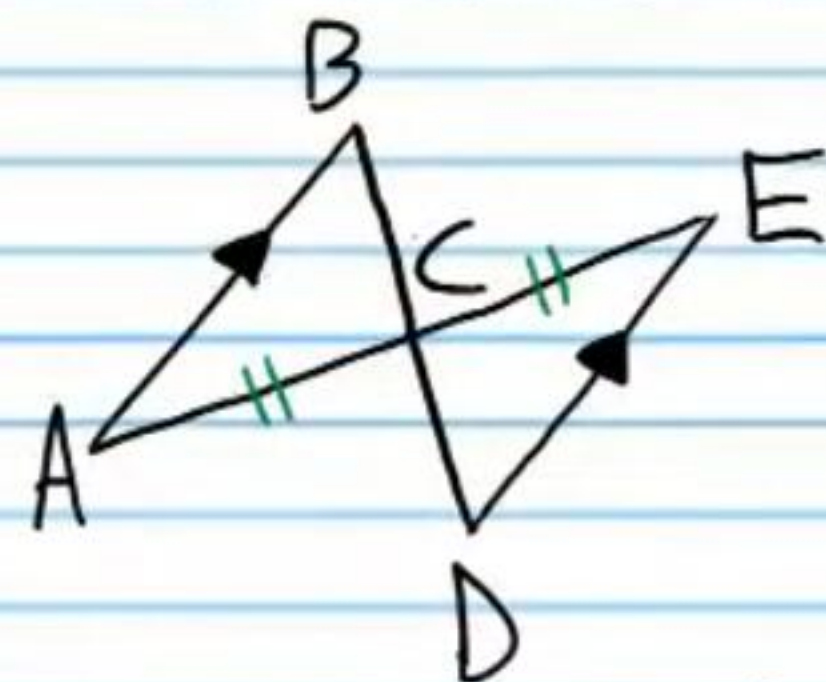
Show that the following statements are true using two-column proofs. You might not use the same number of steps written in Roman numerals. You may use more or less.

②① If  $x=4$ , then point B bisects  $\overline{AC}$ .



| Statement                       | Reason               |
|---------------------------------|----------------------|
| I) $x=4$                        | Given.               |
| II) $AB=3+x$ and $BC=19-3x$     | Given (diagram)      |
| III) $AB=3+4$ and $BC=19-3(4)$  | Subs. Prop of =.     |
| IV) $AB=7$ and $BC=7$           | Simplification       |
| V) $AB=BC$                      | Trans. Prop of =.    |
| VI) B bisects $\overline{AC}$ . | Def. of seg bisector |

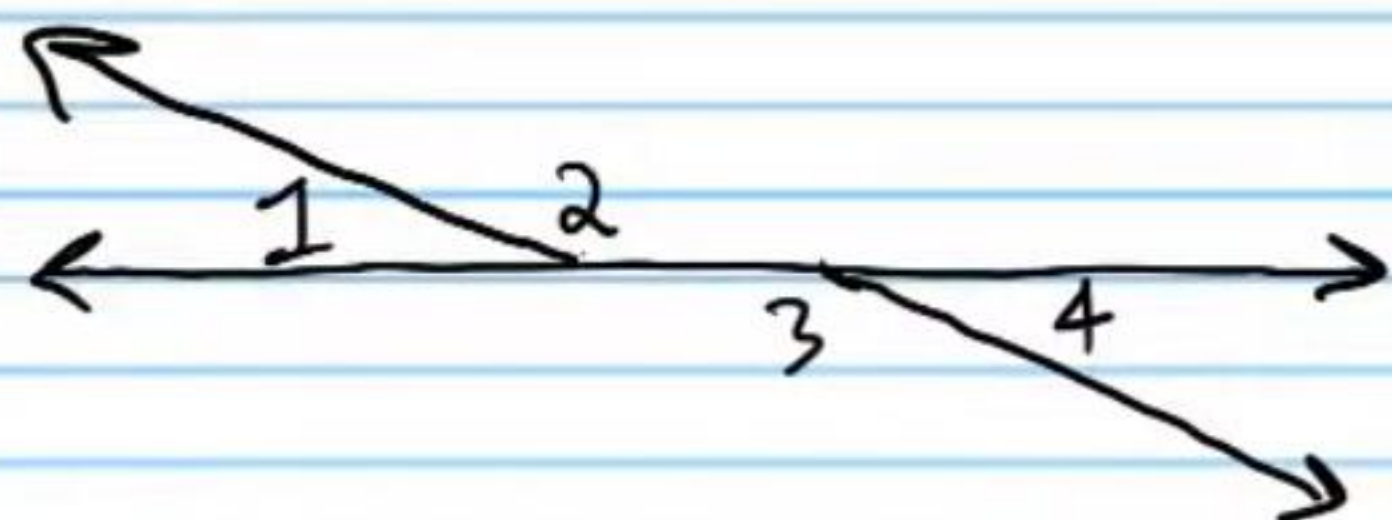
②① If  $\overline{AB} \parallel \overline{ED}$  below, and  $\overline{AC} \cong \overline{CE}$ , show that C is the midpoint of  $\overline{BD}$  with a two-column proof.



| Statement                                                     | Reason           |
|---------------------------------------------------------------|------------------|
| I) $\overline{AB} \parallel \overline{ED}$                    | Given.           |
| II) $\angle A \cong \angle E$ and $\angle B \cong \angle D$ . | Alt Angles       |
| III) $\overline{AC} \cong \overline{CE}$                      | Given.           |
| IV) $\triangle ABC \cong \triangle EDC$                       | AAS              |
| V) $\overline{BC} \cong \overline{CD}$                        | CPCTC            |
| VI) C is the midpoint of $\overline{BD}$ .                    | Def. of midpoint |
| VII)                                                          |                  |

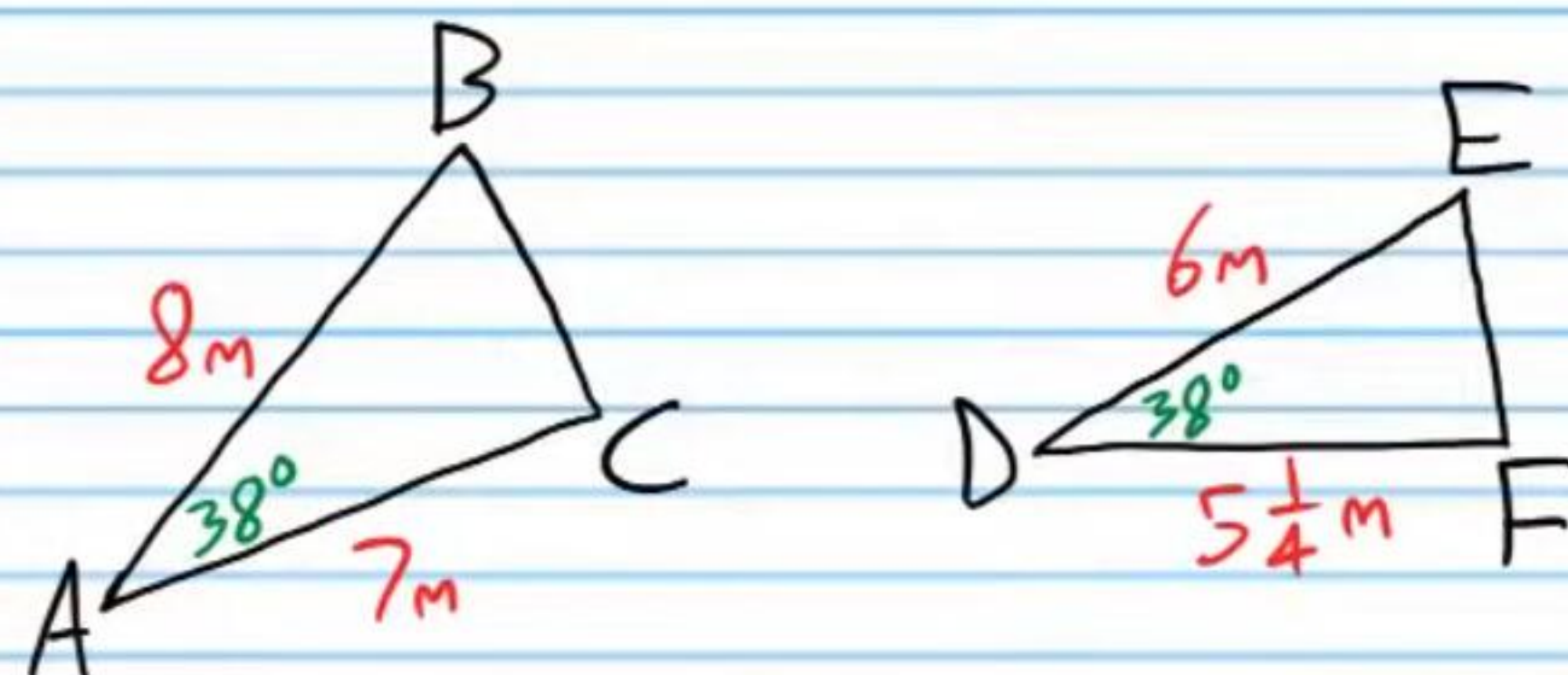


② Use the Congruent Supplements Theorem to show that if  $\angle 2$  and  $\angle 4$  are supplementary, then  $m\angle 2 = m\angle 3$ .



| Statement                                         | Reason                       |
|---------------------------------------------------|------------------------------|
| I) $\angle 2$ and $\angle 4$ are supplementary.   | Given                        |
| II) $\angle 3$ and $\angle 4$ form a linear pair. | Given (diagram)              |
| III) $\angle 3$ and $\angle 4$ are supplementary. | Linear Pair Post.            |
| IV) $\angle 2 \cong \angle 3$                     | $\cong$ Supp. Thm            |
| V) $m\angle 2 = m\angle 3$                        | Def. of $\cong$ $\angle$ 's. |

③  $\triangle ABC$  and  $\triangle DEF$  are similar.



| Statement                                                                      | Reason                       |
|--------------------------------------------------------------------------------|------------------------------|
| I) $AB = 8, AC = 7, DE = 6, \& DF = 5\frac{1}{4}$                              | Given (diagram)              |
| II) $\frac{AB}{DE} = \frac{8}{6}$ and $\frac{AC}{DF} = \frac{7}{5\frac{1}{4}}$ | Subs. Prop of =.             |
| III) $\frac{AB}{DE} = 1.\bar{3}$ and $\frac{AC}{DF} = 1.\bar{3}$               | Simplification               |
| IV) $\frac{AB}{DE} = \frac{AC}{DF}$                                            | Trans. Prop of =.            |
| V) $m\angle A = 38^\circ$ and $m\angle D = 38^\circ$                           | Given (diagram)              |
| VI) $m\angle A = m\angle D$                                                    | Trans. Prop of =.            |
| VII) $\angle A \cong \angle D$                                                 | Def. of $\cong$ $\angle$ 's. |
| VIII) $\triangle ABC \sim \triangle DEF$                                       | SAS                          |



②④ Show that the quadrilateral below is a square.

$$\begin{aligned} D_{AB} &= \sqrt{(1-2)^2 + (3-(-2))^2} \\ &= \sqrt{(-1)^2 + (3+2)^2} \\ &= \sqrt{1+5^2} \\ &= \sqrt{1+25} \\ &= \sqrt{26} \end{aligned}$$

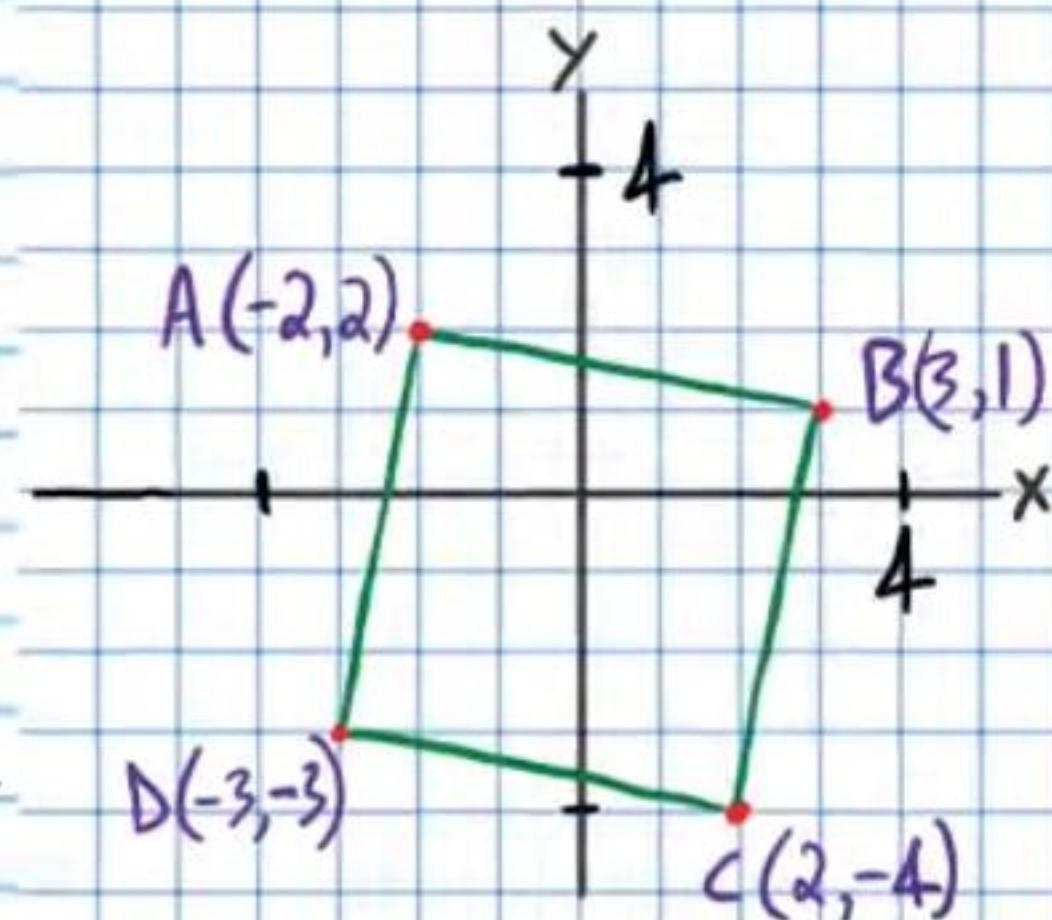
$$\begin{aligned} D_{BC} &= \sqrt{(1-(-4))^2 + (3-2)^2} \\ &= \sqrt{26} \end{aligned}$$

$$\begin{aligned} D_{CD} &= \sqrt{(-4-(-3))^2 + (2-(-3))^2} \\ &= \sqrt{26} \end{aligned}$$

$$\begin{aligned} D_{AD} &= \sqrt{(2-(-3))^2 + (-2-(-3))^2} \\ &= \sqrt{26} \end{aligned}$$

Therefore, all sides are congruent.

Therefore, ABCD is a square.



$$m_{AB} = \frac{2-1}{-2-3} = -\frac{1}{5}$$

$$m_{BC} = \frac{1-(-4)}{3-2} = 5$$

Therefore,  $\angle B$  is a right angle.

$$m_{CD} = \frac{-4-(-3)}{2-(-3)} = -\frac{1}{5}$$

Therefore,  $\angle C$  is a right angle.

$$m_{AD} = \frac{2-(-3)}{-2-(-3)} = 5$$

Therefore,  $\angle D$  &  $\angle A$  are right angles

②⑤ Find the coordinates of the centroid of the triangle below.

Median 1

$$y = 3$$

Median 2

$$M_{JK} \left( \frac{-2+2}{2}, \frac{5+3}{2} \right)$$

$$M_{JK}(0, 4)$$

$$A(0, 4) \leftrightarrow K(-2, 1)$$

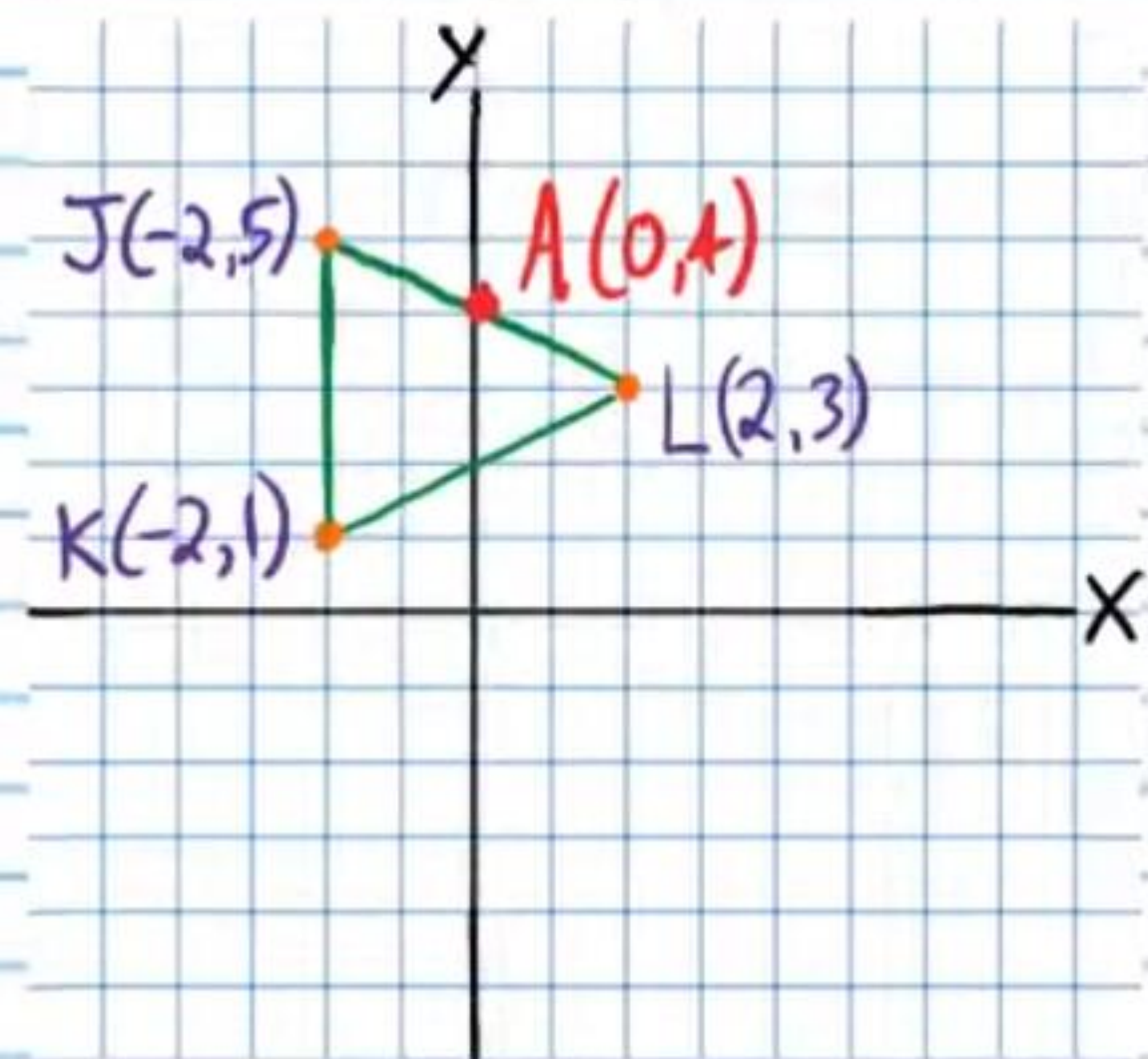
$$m_{AK} = \frac{1-4}{-2-0} = \frac{-3}{-2} = \frac{3}{2}$$

$$y - y_p = m(x - x_p)$$

$$y - 4 = \frac{3}{2}(x - 0)$$

$$y - 4 = \frac{3}{2}x$$

$$y = \frac{3}{2}x + 4$$



$$y = y$$

$$3 = \frac{3}{2}x + 4$$

$$-1 = \frac{3}{2}x$$

$$-2 = 3x$$

$$-\frac{2}{3} = x$$

$$\left( -\frac{2}{3}, 3 \right)$$